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Unit – 5 : Session – 10 : SLO – 2 | ARM Addressing Modes

■ I. Multiple Choice

1. C) Pre-indexed — the offset is applied to the base register before the memory access.
2. B) LDR R0, =value — the pseudo-instruction loads a constant from the literal pool.
3. B) Access memory — post-indexed accesses memory at the old base first, then updates the register.

■ II. Fill in the Blanks

1. register — register-offset addressing uses a register operand as the offset.
2. literal — literal addressing loads constants from the literal pool.
3. after — post-indexed addressing updates the base register after accessing memory.

■ III. True or False

1. TRUE.
2. FALSE — LDR/STR do support scaled index addressing (e.g. [R1, R2, LSL #2]).
3. TRUE — the literal pool is accessed PC-relative.

■ IV. Match the Following

- A) Immediate Offset — 3) Uses immediate displacement.
B) Pre-indexed — 2) Offset added before access.
C) Post-indexed — 4) Base updated after access.
D) Literal Addressing — 1) Uses PC-relative constant.

■ V. Numerical Answers

1. 0x200C — $0x2000 + 12 = 0x200C$.
2. 0x1010 — $R2 \text{ LSL } \#2 = 4 \times 4 = 16 = 0x10$; $0x1000 + 0x10 = 0x1010$.
3. 0x3004 — post-indexed: memory is accessed at 0x3000, then $R1 = 0x3000 + 4 = 0x3004$.